Precise Attribution of Model Interventions: Attribution Sets, Mechanism Reach, Path Defects and Gluing

A companion to *The Three-Level Posterior Tower*

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Abstract

A language model is trained on a corpus, extended with a second corpus, edited directly in its weights, and tuned with reinforcement learning from human feedback. Each step changes some outputs and leaves others alone, and new outputs appear. We ask three questions about such a history: what explains each observed change, how precisely the data determine that explanation, and whether the answer depends on the order of the steps. We answer them with three notions. An *attribution set* collects every decomposition of an observed output distribution into declared sources and an unmodeled remainder; it is a polytope, and its emptiness, its range of unmodeled weight and its certificates are computed by linear programming. *Precision* is the width of the attribution set; it is monotone along a ladder from exact sources to sources known by their supports to a positive budget for unmodeled sources, and any slack in that budget forces imprecision. *Mechanism reach* declares what an intervention type may change: we prove that the reach set of KL-regularized reinforcement learning with a bounded reward range is a polytope, so that attributing a change to it is a linear program with an exact certificate, whereas a fixed reward with unknown temperature gives a curve, and an unconstrained edit can explain anything. Finally, interventions are compared quantitatively: comparisons compose by the triangle inequality and by a Lipschitz rule, with exact Lipschitz constants for mixing and bounds for tilts, and a weight-level intervention may have no Lipschitz constant on observed outputs at all. Order and loop defects measure path dependence, and coarse evaluations certify fine differences. The three kinds of uncertainty involved (who could have produced an output, what a mechanism can produce, and what observation reveals about an intervention) arise from three maps that fail to be injective; we make their interaction precise through the observable reach of an intervention and a soundness condition for mechanism classes, and show that the parameters of a reachable tilt are identified exactly up to the slack in the reward range. We then supply the glue between local structures. Markov kernels are the 1-cells; they transport attribution sets and can only lose precision. Couplings are 2-cells that witness how two 1-cells differ. Their vertical composition is the gluing lemma and is strictly associative, but the natural whiskerings are not functorial, and the strict 2-categorical structure is a graded metric shadow. For several contexts, two compatible contexts always glue, and that gluing is vertical composition of witnesses. For a general cover, the contextual fraction of Abramsky, Barbosa and Mansfield equals our minimal unmodeled weight, and the best approximate global state is computed by a linear program; for the Popescu–Rohrlich box it is exactly 1/4 away from some

*Code and checks: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/precise_attribution (`check_attribution.py`). Verification scripts: `provenance.py`, `check_attribution.py`, `check_glue.py`, `provenance_demo.py`; see appendix A.

context. Gluing at the second level of the tower retains correlations that gluing flattened states loses. Finally, the simplex of output distributions is a moment polytope whose Lagrangian fibres are what observation forgets, and we state necessary conditions for a Fukaya-type realization; a companion theory paper has since decided parts of two of them. A synthetic four-stage experiment produces a full provenance report. Every theorem is checked in exact arithmetic or with exactly verified linear programming certificates.

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1 Introduction

Consider the life of a model. Corpus C_1 produces model M_1. Adding corpus C_2 produces M_2, which changes some outputs, leaves others unchanged, and makes outcomes probable that were not before. An engineer then edits weights directly, producing M_3, and reinforcement learning from human feedback (RLHF) produces M_4. For such a history we want answers to three questions.

(Q1) *Attribution.* Which declared source or intervention explains each observed change, and what is left unexplained?

(Q2) *Precision.* How tightly do the observations determine that explanation?

(Q3) *Path dependence.* Does the result depend on the order of interventions, and does a sequence that should cancel actually cancel?

Three kinds of uncertainty. Each question concerns a different map that fails to be injective, and the paper is organized around them.

Question	Object	Source of the uncertainty	Measured and certified by
Who could have produced p?	attribution set $W_{\bar{\mu}}(p)$ (§3)	inverse image of p under the mixing map from decompositions to outputs	widths (definition 3.4); dual certificate (theorem 3.2)
What can mechanism T produce?	reach set $\mathrm{Reach}_T(p)$ (§4)	image of the mechanism's parameters, which may themselves not be identified	residual weight and certificate (theorem 4.9); parameter widths (proposition 4.7)
What can we observe about T?	$\Phi \circ T$ (§5)	quotient by Φ, which merges models with equal probe outputs	comparisons, Lipschitz constants and defects (theorem 5.5)

The second and third interact: an intervention acts on model states, so what can be seen of it from observed outputs is set-valued even when the intervention is deterministic. The *observable reach* and the *soundness* of a mechanism class (definition 4.11 and proposition 4.12) make this precise, and they separate two claims that are easily conflated: that a change is not a tilt, which is a fact about outputs, and that a change was not produced by RLHF, which also requires the tilt class to be sound for the procedure actually run.

Architecture. The companion paper [9] develops the *local* structure: an ambient probability monad D, and over each outcome base B a context comonad, a mixed distributive law, admissibility data and an obstruction tower. Its §5 establishes identifiability and contamination results for a single output distribution. The present paper builds the *transport* half of the glue between local structures: interventions as maps between model states, and quantitative comparisons between interventions. Section 7 makes the glue explicit: kernels as 1-cells, couplings as 2-cells with a strict graded shadow, and assembly of several contexts into a global state, exactly or approximately. Section 8 says how the layers fit together, which laws are strict, and where symplectic geometry and Fukaya categories could enter.

Contributions.

1. Attribution sets for declared sources known exactly or by their supports, with an unmodeled budget; their structure, emptiness criterion and certificates (theorem 3.2).

2. Precision as width, a monotone precision ladder separating structural, informational and unmodeled imprecision (theorem 3.5), and the fact that budget slack forces imprecision (proposition 3.6).

3. Mechanism classes and a change-decomposition theorem with certificates (theorem 4.9); the reach set of KL-regularized RLHF with bounded rewards is a polytope (theorem 4.4), while a fixed reward with unknown temperature is not convex (proposition 4.6).

4. Quantitative comparison of interventions with vertical and horizontal composition rules and an accumulation bound (theorem 5.5); exact or bounded Lipschitz constants for mixing, tilts and coarsening, and the absence of any observable Lipschitz constant for some weight-level interventions (propositions 5.3 and 5.4); order and loop defects; monotonicity under coarsening (proposition 5.9).

5. Observable reach and soundness of mechanism classes, with an enlargement that keeps the RLHF class sound for approximate optimization (proposition 4.12 and corollary 4.13); identifiability of tilt parameters up to the slack in the reward range (proposition 4.7).

6. A synthetic four-stage experiment with a complete provenance report (section 6).

7. The glue: kernel 1-cells transport attribution and only lose precision (proposition 7.3); coupling 2-cells form hom-categories under gluing, with explicit failures of functorial whiskering (theorem 7.6); the graded shadow is a strict 2-category (theorem 7.8).

8. Assembly: two-context gluing is vertical composition of witnesses (proposition 7.11); the contextual fraction equals the minimal unmodeled weight (theorem 7.12), which also detects prompt-dependent attribution (example 7.13); bounds for approximate gluing with the exact value $\frac{1}{4}$ for the PR box (proposition 7.15); gluing at level 2 (proposition 7.16).

9. An architecture that separates what is strict from what holds up to error, and a geometric placement with checkable conditions for a Fukaya-type realization (section 8).

Several ingredients are known in other fields; section 9 states precisely which, and what is new here.

What is not claimed. We do not extract hypotheses from weights: declared sources must come from reference models, checkpoints or ablations. All guarantees concern a finite probe set. Attribution relative to a declared mechanism is not identification of a cause; causal statements require running the compared paths. We treat output distributions as known; finite-sample precision is an open problem (section 10).

2 Model states and observation

Definition 2.1 (Observable states)**.** Fix a finite set X of *probes* (prompts or evaluation contexts) and, for each $x \in X$, a finite set B_x of outcomes, which may be coarse classes of completions. The *observable state space* is $\mathcal{P} = \prod_{x \in X} \mathsf{D}(B_x)$, with

$$d(p, q) = \max_{x \in X} \mathrm{TV}(p_x, q_x), \qquad \mathrm{TV}(p_x, q_x) = \tfrac{1}{2} \textstyle\sum_b |p_x(b) - q_x(b)|.$$

Definition 2.2 (Model states and interventions)**.** A set $\mathcal{W}$ of *model states* (for instance weight vectors) carries an *observation map* $\Phi \colon \mathcal{W} \to \mathcal{P}$ sending a model to its output distributions on the probes. An *intervention* is a map $T \colon \mathcal{W} \to \mathcal{W}$; its *observable effect* on M is the pair $(\Phi M, \Phi TM)$.

Remark 2.3 (Observation is not faithful)*.* Different models can have the same outputs on every probe, and the same intervention can then produce different outputs. This is why statements about interventions on weights cannot in general be reduced to statements about observable states (proposition 5.4).

Definition 2.4 (New outcomes)**.** A softmax output assigns positive probability to every outcome, so "a new outcome appears" is made precise by a threshold $\varepsilon > 0$: outcome b *appears at level* ε at probe x under $M \to M'$ if $(\Phi M)_x(b) \le \varepsilon < (\Phi M')_x(b)$.

From section 3 to section 4 we fix a probe, write B for its finite outcome set and $p \in \mathsf{D}(B)$ for an observed output distribution. All statements apply probe by probe.

3 Attribution sets and precision

Definition 3.1 (Declared sources and attribution sets)**.** A *declared source* is either *exact*, a distribution $A_i \in \mathsf{D}(B)$, or *known by its support*, a set $S_i \subseteq B$. An *unmodeled budget* is a number $\bar{\mu} \in [0,1]$. A *decomposition* of p is a family of nonnegative measures $s_1, \dots, s_k$ on B, the *contributions*, such that

(a) $s_i = \lambda_i A_i$ for some $\lambda_i \geq 0$ if source i is exact, and $\operatorname{supp} s_i \subseteq S_i$ if source i is known by its support, in which case $\lambda_i = \operatorname{mass}(s_i)$;

(b) $\sum_i s_i \leq p$ pointwise, and $\mu := 1 - \sum_i \lambda_i \leq \bar{\mu}$.

The *unmodeled part* is $p - \sum_i s_i = \mu U$ with $U \in \mathsf{D}(B)$ when $\mu > 0$. The *attribution set* $W_{\bar{\mu}}(p)$ is the set of all decompositions. The *minimal unmodeled weight* $\mu_{\min}(p)$ is the least μ over decompositions with $\bar{\mu} = 1$.

The model is Huber's contamination model [23] with declared sources. A decomposition determines U, so the attribution set records exactly the explanations compatible with p.

Theorem 3.2 (Structure of attribution sets)**.**

(i) $W_{\bar{\mu}}(p)$ *is a polytope, cut out by linear constraints on the contributions.*

(ii) $W_{\bar{\mu}}(p) \neq \emptyset$ *if and only if* $\mu_{\min}(p) \leq \bar{\mu}$.

(iii) If $W_{\bar{\mu}}(p) \neq \emptyset$*, the unmodeled weights it contains form the interval* $[\mu_{\min}(p), \bar{\mu}]$.

(iv) (Certificate) Suppose $y \in \mathbb{R}^B_{\geq 0}$ *satisfies* $\langle y, a \rangle \geq 1$ *for every admissible version* a *of every source, that is* $\langle y, A_i \rangle \geq 1$ *for exact sources and* $y_b \geq 1$ *for* $b \in S_i$ *for support-known sources. Then every decomposition has* $\mu \geq 1 - \langle y, p \rangle$*, and the best such bound equals* $\mu_{\min}(p)$.

Proof. (i) Constraints (a) and (b) are linear in the variables λ_i (exact sources) and s_i (support-known sources), with $s_i \geq 0$, and the feasible set is bounded since $\sum_i s_i \leq p$. (ii) and (iii) If a decomposition has weight vector λ, then so does $t\lambda$ (with scaled contributions) for every $t \in [0,1]$, which raises μ continuously up to 1. Hence the attainable μ form $[\mu_{\min}(p), 1]$, and the budget cuts this to $[\mu_{\min}(p), \bar{\mu}]$. (iv) Each contribution is a nonnegative multiple $\lambda_i a$ of an admissible version a of its source, so $\langle y, \sum_i s_i \rangle \geq \sum_i \lambda_i = 1 - \mu$, while $\langle y, \sum_i s_i \rangle \leq \langle y, p \rangle$. Optimality of the best bound is linear programming duality [40]: $1 - \mu_{\min}(p)$ is the value of $\max\{\sum_i \lambda_i\}$ over (a), (b), and its dual is $\min\{\langle y, p \rangle\}$ under the stated constraints on y. □

Remark 3.3 (Mixture proportion estimation). With a single exact source A, $\mu_{\min}(p) = 1 - \kappa^*(p \,|\, A)$, where $\kappa^*(p \,|\, A) = \min_{b \in \operatorname{supp} A} p(b)/A(b)$ is the maximal proportion of A in p studied in mixture proportion estimation [10, 41]. A positive budget $\bar{\mu}$ plays the role of the bound on the error probability in contaminated sampling [22].

Definition 3.4 (Precision)**.** For a nonempty attribution set, the *width* of source i is $w_i = \max \lambda_i - \min \lambda_i$ over $W_{\bar{\mu}}(p)$, and the width of the unmodeled weight is $w_\mu = \max \mu - \min \mu$. For an outcome b with $p(b) > 0$, the *observation-level attribution* $\pi_i(b) = s_i(b)/p(b)$ is the probability that an observation of b came from source i; its width is defined likewise. The attribution is *exact* if all widths vanish. All widths are values of linear programs over $W_{\bar{\mu}}(p)$.

Theorem 3.5 (Monotone precision ladder)**.** *Enlarging the admissible versions of any source, or raising the budget, can only enlarge the attribution set; hence all widths are nondecreasing. In particular, writing $\mu_{\min}^{\mathrm{ex}}$ for the minimal unmodeled weight with exact sources, the attribution sets*

$$W^{(1)} = W^{\mathrm{exact}}_{\mu_{\min}^{\mathrm{ex}}}(p) \;\subseteq\; W^{(2)} = W^{\mathrm{support}}_{\mu_{\min}^{\mathrm{ex}}}(p) \;\subseteq\; W^{(3)} = W^{\mathrm{support}}_{\bar{\mu}}(p) \qquad (\bar{\mu} \geq \mu_{\min}^{\mathrm{ex}})$$

are nonempty and nested, and each width satisfies $w^{(1)} \leq w^{(2)} \leq w^{(3)}$.

Proof. Every decomposition with exact sources is one with the corresponding supports, since $\mathrm{supp}(\lambda_i A_i) \subseteq \mathrm{supp}\, A_i$; and every decomposition within budget $\bar{\mu}$ is one within any larger budget. The width of a coordinate over a larger set is at least its width over a smaller set. $W^{(1)}$ is nonempty by theorem 3.2(ii). □

We call the increments $w^{(1)}$, $w^{(2)} - w^{(1)}$ and $w^{(3)} - w^{(2)}$ the *structural*, *informational* and *unmodeled* imprecision. They are nonnegative by theorem 3.5; we do not claim that they combine additively across sources.

Proposition 3.6 (Budget slack forces imprecision)**.** *If $\mu_{\min}(p) < \bar{\mu}$ and $\mu_{\min}(p) < 1$, then $w_\mu = \bar{\mu} - \mu_{\min}(p) > 0$, and every source whose maximal weight is positive has positive width.*

Proof. The first claim is theorem 3.2(iii). Take a decomposition λ with $\mu = \mu_{\min}(p)$. For $t = (1-\bar{\mu})/(1-\mu_{\min}(p)) < 1$, the scaled family $t\lambda$ is again a decomposition, since its unmodeled weight is $\bar{\mu}$. If $\lambda_i > 0$, the two decompositions give different values of the i-th weight. If $\lambda_i = 0$, this value differs from the positive maximal weight. Either way $w_i > 0$. □

So precision and robustness trade off: a budget for unmodeled sources protects against rejecting the declared model, and every unit of slack in that budget is paid for in width.

Proposition 3.7 (When attribution is exact)**.** *With exact sources and budget $\mu_{\min}(p) = 0$, the attribution is exact for every p in $\mathrm{conv}(A_1, \ldots, A_k)$ if and only if the sources are affinely independent. For any sources, if the exclusive region $R_i = \{b : A_i(b) > 0,\ A_j(b) = 0\ (j \neq i)\}$ has $A_i(R_i) > 0$, then $\lambda_i = p(R_i)/A_i(R_i)$ in every decomposition, and $\pi_i(b) = \lambda_i A_i(b)/p(b)$.*

Proof. This is [9, Prop. 5.7]. □

4 Mechanism classes and attribution of change

Definition 4.1 (Mechanism class)**.** A *mechanism class* for an intervention type assigns to each $p \in \mathsf{D}(B)$ a set $\mathrm{Reach}(p) \subseteq \mathsf{D}(B)$ of output distributions the intervention may produce from p. A change $p \to p'$ is *reachable* if $p' \in \mathrm{Reach}(p)$.

Example 4.2 (Three polyhedral classes)**.**

(a) *Source expansion* (adding a corpus): $\mathrm{Reach}(p) = \mathrm{conv}(\mathcal{A} \cup \mathcal{N})$, where $\mathcal{A}$ are the old sources and $\mathcal{N}$ the new ones.

(b) *Locality claim* for an edit with declared scope $X_0 \subseteq X$ and tolerance τ: at a probe outside X_0, $\mathrm{Reach}(p) = \{q : \mathrm{TV}(q, p) \leq \tau\}$.

(c) *Unconstrained edit*: $\mathrm{Reach}(p) = \mathsf{D}(B)$.

Proposition 4.3 (KL-regularized reinforcement learning is an exponential tilt)**.** *For a reward* $r\colon B \to \mathbb{R}$ *and* $\beta > 0$*, the maximizer of* $\mathbb{E}_q[r] - \beta\,\mathrm{KL}(q \,\|\, p)$ *over* $q \in \mathsf{D}(B)$ *is*

$$T_r(p)(b) = \frac{p(b)\, e^{r(b)/\beta}}{\sum_c p(c)\, e^{r(c)/\beta}}.$$

Proof. For $q \ll p$, $\mathbb{E}_q[r] - \beta\,\mathrm{KL}(q\|p) = \beta \log Z - \beta\,\mathrm{KL}(q\|T_r(p))$ with $Z = \sum_c p(c)e^{r(c)/\beta}$, which is maximal exactly at $q = T_r(p)$; and $q \not\ll p$ gives $-\infty$. This is the standard Gibbs variational principle, used for RLHF in [28, 37]. □

Writing $w(b) = e^{(r(b)-r_{\min})/\beta}$, a reward with values in $[r_{\min}, r_{\max}]$ gives weights in $[1, K]$ with $K = e^{(r_{\max}-r_{\min})/\beta}$, and $T_r(p)$ depends only on w. The tilt is a reweighting by a likelihood, the operation called conditioning in [9, §4]; it is not conditioning on an observed event. It preserves support exactly, $\operatorname{supp} T_r(p) = \operatorname{supp} p$, but it can raise a small probability by a factor up to K.

Theorem 4.4 (The tilt reach set is a polytope)**.** *Let* $\mathrm{Reach}_K(p) = \{T_r(p) : r(B) \subseteq [r_{\min}, r_{\max}]\}$ *with* K *as above. Then*

$$\mathrm{Reach}_K(p) = \big\{q \in \mathsf{D}(B) :\ \operatorname{supp} q = \operatorname{supp} p,\quad q_b\, p_c \le K\, q_c\, p_b \ \ \textit{for all } b, c \in \operatorname{supp} p\big\},$$

a convex polytope. Equivalently $q \in \mathrm{Reach}_K(p)$ *iff* $\operatorname{supp} q = \operatorname{supp} p$ *and*

$$K^*(p, q) = \frac{\max_{b \in \operatorname{supp} p} q_b/p_b}{\min_{b \in \operatorname{supp} p} q_b/p_b} \le K,$$

so $\beta \log K^*(p, q)$ *is the least reward range that explains the change.*

Proof. If $q = T_r(p)$, then $q_b/p_b = w_b/Z$ for $b \in \operatorname{supp} p$, so the ratios lie within a factor K of each other, which is the stated inequality. Conversely, if q satisfies the inequalities and has support $\operatorname{supp} p$, put $w_b = (q_b/p_b)/\min_c(q_c/p_c) \in [1, K]$ for $b \in \operatorname{supp} p$ and $w_b = 1$ elsewhere; then $T_w(p) = q$. The inequalities are linear in q for fixed p; together with $\sum_b q_b = 1$ and $q_b = 0$ off $\operatorname{supp} p$ they define a polytope. A point satisfying them cannot vanish anywhere on $\operatorname{supp} p$: if $q_b = 0$ then $q_c\, p_b \le K\, q_b\, p_c = 0$ for all c, contradicting $\sum q = 1$. So the constraint $\operatorname{supp} q = \operatorname{supp} p$ is implied, and the set is closed. □

Remark 4.5 (Density ratio classes). The set in theorem 4.4 is the interval of measures $\{q : p \le c\,q \le Kp$ for some $c > 0\}$ of DeRobertis and Hartigan [17], known in robust Bayesian analysis as a density ratio class. Theorem 4.4 therefore identifies the reach of KL-regularized RLHF with bounded rewards as a classical object; what it adds is that identification, and the reading of $\beta \log K^*$ as the least reward range that explains an observed change.

Proposition 4.6 (Fixed reward, unknown temperature)**.** *For a fixed reward* r*, the set* $\{T_{r/\beta'}(p) : \beta' > 0\}$ *is a curve, and in general not convex: for* p *uniform on three outcomes and* $r = (0, 1, 2)$*, the midpoint of the points with* $1/\beta' = 1$ *and* $1/\beta' = 3$ *is not on the curve.*

Proof. On the curve, $\log(q_b/p_b) - \log(q_0/p_0)$ is proportional to $r_b - r_0$, so $\log(q_2/q_0) = 2\log(q_1/q_0)$ for this p and r; the midpoint violates this identity, as checked in appendix A. □

So the geometry of a reach set depends on what is assumed about the mechanism: a bounded reward range gives a polytope, a fixed reward gives a curve, and no assumption gives everything.

A reachable change can still leave the mechanism's parameters undetermined. For tilts this is measured exactly.

Proposition 4.7 (Identifiability of tilt parameters)**.** *Fix $\beta > 0$ and a reward range $[r_{\min}, r_{\max}]$ of length Δ_r, and let $q \in \mathrm{Reach}_K(p)$ with $K = e^{\Delta_r/\beta}$. Put $\ell_b = \beta \log(q_b/p_b)$ for $b \in \operatorname{supp} p$. The rewards r with values in $[r_{\min}, r_{\max}]$ and $T_r(p) = q$ are exactly*

$$r_b = \ell_b + c \quad (b \in \operatorname{supp} p), \qquad c \in \big[r_{\min} - \min_b \ell_b,\ r_{\max} - \max_b \ell_b\big],$$

with $r_b \in [r_{\min}, r_{\max}]$ arbitrary for $b \notin \operatorname{supp} p$. Hence on $\operatorname{supp} p$ every reward value has width $\Delta_r - \beta \log K^*(p, q)$, reward differences are identified exactly, and the reward is identified exactly on $\operatorname{supp} p$ if and only if $\beta \log K^*(p, q) = \Delta_r$. Only r/β enters T_r, so without a known β only the ratios of reward differences are identified.

Proof. $T_r(p) = q$ holds iff $q_b/p_b = e^{r_b/\beta}/Z$ on $\operatorname{supp} p$, that is, iff $r_b - \beta \log(q_b/p_b)$ is constant on $\operatorname{supp} p$; outcomes outside the support do not affect $T_r(p)$. The range constraint on $\operatorname{supp} p$ is $r_{\min} \le \ell_b + c \le r_{\max}$ for all b, which is the stated interval for c; its length is $\Delta_r - (\max_b \ell_b - \min_b \ell_b) = \Delta_r - \beta \log K^*(p, q)$. □

This mirrors proposition 3.6: slack in the assumed reward range costs precision about the reward, exactly as slack in the unmodeled budget costs precision about the sources.

Definition 4.8 (Residual of a change)**.** Let $\operatorname{cone}\mathrm{Reach}(p) = \{tq : t \ge 0,\ q \in \mathrm{Reach}(p)\}$. An *explained part* of a change $p \to p'$ is a measure $s \in \operatorname{cone}\mathrm{Reach}(p)$ with $s \le p'$; the remainder $p' - s$ is the *residual*. The *residual weight* of the change is

$$\mu^T_{\min}(p, p') = 1 - \max\{\mathrm{mass}(s) :\ s \in \operatorname{cone}\mathrm{Reach}(p),\ s \le p'\}.$$

Theorem 4.9 (Change decomposition)**.** *Let $\mathrm{Reach}(p)$ be closed and convex.*

(i) The maximum defining $\mu^T_{\min}(p, p')$ is attained, and $\mu^T_{\min}(p, p') = 0$ if and only if $p' \in \mathrm{Reach}(p)$.

(ii) (Certificate) If $y \ge 0$ satisfies $\langle y, q\rangle \ge 1$ for all $q \in \mathrm{Reach}(p)$, then $\mu^T_{\min}(p, p') \ge 1 - \langle y, p'\rangle$. If $\operatorname{cone}\mathrm{Reach}(p) = \{s \ge 0 : Gs \le 0\}$ is polyhedral, the best such bound is attained by a pair $y \ge 0$, $\nu \ge 0$ with $y + G^\top \nu \ge \mathbf{1}$, and it equals $\mu^T_{\min}(p, p')$.

(iii) (Precision of change attribution) The optimal explained parts form a polytope when the cone is polyhedral, and the range of each $s(b)$ over it is given by linear programs.

Proof. (i) $\mathrm{Reach}(p) \subseteq \mathsf{D}(B)$ is compact and avoids 0, so its cone is closed; intersected with $\{0 \le s \le p'\}$ it is compact, and mass is continuous. If the maximum is 1, then $s \le p'$ and $\mathrm{mass}(s) = \mathrm{mass}(p')$ force $s = p'$, so $p' \in \operatorname{cone}\mathrm{Reach}(p)$ with mass one, that is, $p' \in \mathrm{Reach}(p)$. (ii) For $s = tq$ with $q \in \mathrm{Reach}(p)$, $\mathrm{mass}(s) = t \le t\langle y, q\rangle = \langle y, s\rangle \le \langle y, p'\rangle$. In the polyhedral case, $\max\{\mathbf{1}^\top s : Gs \le 0,\ s \le p',\ s \ge 0\}$ has dual $\min\{\langle y, p'\rangle : y + G^\top \nu \ge \mathbf{1},\ y, \nu \ge 0\}$, and strong duality holds [40]; every dual-feasible pair satisfies $\langle y, s\rangle \ge \mathbf{1}^\top s - \langle \nu, Gs\rangle \ge \mathbf{1}^\top s$ on the cone. (iii) The optimal set is a face of a polytope. □

Remark 4.10 (What residuals mean). A positive residual weight says that at least this much of the observed change is not produced by any intervention of the declared type. It does not say what produced it. For the tilt class the cone is $\{s \ge 0 :\ s_b p_c \le K s_c p_b,\ s = 0 \text{ off } \operatorname{supp} p\}$, so attributing a change to RLHF is a linear program with an exact certificate. For an unconstrained edit the residual is always zero: an edit can be tested only through a claim about it, such as locality.

Mechanism classes are defined on observed outputs, while real interventions act on model states. The link between the two is the following.

Definition 4.11 (Observable reach and soundness)**.** For an intervention $T\colon \mathcal{W} \to \mathcal{W}$ and an observable state $p \in \mathcal{P}$, the *observable reach* is

$$R_T^{\Phi}(p) = \{\Phi T M : \ M \in \mathcal{W},\ \Phi M = p\}.$$

A mechanism class is *sound* for T if $R_T^{\Phi}(p)$ is contained, probe by probe, in $\mathrm{Reach}_T(p)$ for every p in the image of Φ.

The observable reach can have more than one point even for a deterministic T: in the example of proposition 5.4 it has two. So mechanism classes must be set-valued even for deterministic interventions.

Proposition 4.12 (What a residual certifies)**.** *Let the mechanism class be sound for T, and suppose the change $p \to p'$ has positive residual weight at some probe. Then there is no model M with $\Phi M = p$ and $\Phi T M = p'$: the change was not produced by T from any model with the observed outputs. Without soundness, the residual certifies only that the change lies outside the declared class. A zero residual never certifies that T produced the change.*

Proof. If such an M existed, then $p' \in R_T^{\Phi}(p) \subseteq \mathrm{Reach}_T(p)$ at every probe, so the residual would vanish by theorem 4.9(i). The last sentence holds because other interventions may reach the same outputs. □

The tilt class of theorem 4.4 is sound for an exact optimizer of the KL-regularized objective with rewards in the stated range, by proposition 4.3. Actual RLHF procedures stop short of the optimum, so soundness holds only approximately. The following enlargement restores it while keeping linear programming.

Corollary 4.13 (Enlarged classes)**.** *If a procedure is known to end within total variation ε of the exact tilt, then the enlarged class* $\mathrm{Reach}_K^{\varepsilon}(p) = \{q \in \mathsf{D}(B) : \mathrm{TV}(q, q') \le \varepsilon \text{ for some } q' \in \mathrm{Reach}_K(p)\}$ *is sound for it, and it is again a polytope, so theorem 4.9 applies.*

Proof. Soundness is immediate from the assumption. The set is the projection onto q of the polyhedron $\{(q, q', u) : q' \in \mathrm{Reach}_K(p),\ -u \le q - q' \le u,\ \sum_b u_b \le 2\varepsilon,\ q \in \mathsf{D}(B)\}$, and a bounded projection of a polyhedron is a polytope. □

How large ε is for a given procedure is an empirical question; the corollary only says how to use it once it is known.

5 Comparing interventions: composition and path defects

Definition 5.1 (Quantitative comparison)**.** For interventions F, G and a set $\mathcal{S} \subseteq \mathcal{W}$ of model states,

$$d_{\mathcal{S}}(F, G) = \sup_{M \in \mathcal{S}} d(\Phi F M, \Phi G M).$$

This is a pseudometric on interventions. We write $F \approx_{\varepsilon} G$ on $\mathcal{S}$ if $d_{\mathcal{S}}(F, G) \le \varepsilon$, and call this a *quantitative comparison*.

A quantitative comparison records how far two interventions are apart; it is not by itself a 2-cell in a 2-category, and we do not call it one. Corollary 5.7 identifies the structure in which it is the hom-metric of an enriched category.

Definition 5.2 (Lipschitz on observables)**.** An intervention G is *L-Lipschitz on observables* over $\mathcal{T} \subseteq \mathcal{W}$ if $d(\Phi GM, \Phi GM') \le L\, d(\Phi M, \Phi M')$ for all $M, M' \in \mathcal{T}$. An *output-level mechanism* is a map $T\colon \mathcal{P} \to \mathcal{P}$ acting on observable states; for it the Lipschitz constant is the usual one for d.

Proposition 5.3 (Lipschitz constants of output-level mechanisms)**.**

(i) Mixing with a fixed target, $p \mapsto (1-a)p + a\,t$, is exactly $(1-a)$-Lipschitz.

(ii) A tilt with weights in $[1, K]$ is $\frac{3K-1}{2}$-Lipschitz, and no constant below K works in general.

(iii) Coarsening along any map of outcomes is 1-Lipschitz.

Proof. (i) The difference of images is $(1-a)(p-q)$. (iii) Data processing for total variation. (ii) Let $Z = \langle w, p\rangle \ge 1$ and $Z' = \langle w, q\rangle$. Then $T(p) - T(q) = w \odot (p-q)/Z + w \odot q\,(1/Z - 1/Z')$, so $\|T(p) - T(q)\|_1 \le \big(K\|p-q\|_1 + |Z - Z'|\big)/Z \le K\|p-q\|_1 + |Z - Z'|$, using $\|w \odot q\|_1 = Z'$ and $Z \ge 1$. Since $p - q$ has total mass zero, $|Z - Z'| = |\langle w - \frac{K+1}{2}\mathbf{1}, p - q\rangle| \le \frac{K-1}{2}\|p-q\|_1$. For the lower bound take $p = (\epsilon, 1-\epsilon)$ and $w = (K, 1)$: the first coordinate of $T(p)$ is $K\epsilon/(1 + (K-1)\epsilon)$, whose derivative at $\epsilon = 0$ is K. □

So tilts can amplify differences, which matters for composition.

Proposition 5.4 (No observable Lipschitz constant)**.** *If $\Phi M = \Phi M'$ but $\Phi GM \ne \Phi GM'$, then G is not L-Lipschitz on observables over $\{M, M'\}$ for any L.*

Proof. The left side of the Lipschitz inequality is positive and the right side is zero. □

A weight edit that depends on parts of the model not visible on the probes is of this kind. For such interventions the composition bound below must use a metric on $\mathcal{W}$, and conclusions at the probe level are conditional on that metric.

Theorem 5.5 (Composition of quantitative comparisons)**.**

(i) (Vertical) $d_{\mathcal{S}}(F, H) \le d_{\mathcal{S}}(F, G) + d_{\mathcal{S}}(G, H)$.

(ii) (Horizontal) If G' is L-Lipschitz on observables over $F(\mathcal{S}) \cup F'(\mathcal{S})$, then

$$d_{\mathcal{S}}(GF, G'F') \le d_{F(\mathcal{S})}(G, G') + L\, d_{\mathcal{S}}(F, F').$$

(iii) (Accumulation) For paths $F_n \cdots F_1$ and $G_n \cdots G_1$, let $\mathcal{S}_0 = \mathcal{S}$ and $\mathcal{S}_j = F_j(\mathcal{S}_{j-1})$, put $\varepsilon_j = d_{\mathcal{S}_{j-1}}(F_j, G_j)$, and suppose each G_j is L_j-Lipschitz on observables over the states reached by both paths. Then

$$\delta_j = d_{\mathcal{S}}(F_j \cdots F_1,\, G_j \cdots G_1) \quad \textit{satisfies} \quad \delta_j \le \varepsilon_j + L_j \delta_{j-1},$$

and hence

$$\delta_n \le \sum_{j=1}^{n} \varepsilon_j \prod_{i=j+1}^{n} L_i.$$

Proof. (i) is the triangle inequality for d. (ii) For $M \in \mathcal{S}$,

$$d(\Phi GFM, \Phi G'F'M) \le d(\Phi GFM, \Phi G'FM) + d(\Phi G'FM, \Phi G'F'M);$$

the first term is at most $d_{F(\mathcal{S})}(G, G')$ and the second at most $L\, d(\Phi FM, \Phi F'M)$. (iii) Apply (ii) with $F = F_j$, $G = G_j$ to the states $F_{j-1} \cdots F_1 M$ and $G_{j-1} \cdots G_1 M$, and unroll the recursion. □

Remark 5.6. Composition rules of this kind are standard for behavioral pseudometrics of probabilistic systems [18], in quantitative algebraic reasoning [33] and in sensitivity type systems [38]. Theorem 5.5 is an instance; what is specific here is the observation map Φ, which is not faithful, and the constants for tilts.

Corollary 5.7 (Enriched structure)**.** *Output-level mechanisms on $\mathcal{P}$ with Lipschitz constant at most 1, with $d = d_{\mathcal{P}}$, form a category enriched in pseudometric spaces in Lawvere's sense [29]: $d(\mathrm{id}, \mathrm{id}) = 0$ and $d(GF, G'F') \le d(G, G') + d(F, F')$. Mixings and coarsenings belong to it; tilts do not, since their Lipschitz constant can exceed 1, so in general the structure is the Lipschitz-weighted one of theorem 5.5.*

Definition 5.8 (Order and loop defects)**.** For interventions E and R, the *order defect* on $\mathcal{S}$ is $d_{\mathcal{S}}(RE, ER)$. For a sequence of interventions $T_1, \dots, T_n$, the *loop defect* is $d_{\mathcal{S}}(T_n \cdots T_1, \mathrm{id})$.

A positive order defect certifies that the two orders produce different observable models on the probes; it becomes a causal statement once both orders have actually been run. We call the loop defect a defect rather than a holonomy, since no connection has been defined.

Proposition 5.9 (Coarse evaluation)**.** *For any coarsening g of outcomes, $\mathrm{TV}(g_*p, g_*q) \le \mathrm{TV}(p, q)$. Hence every order, loop or comparison defect measured on a coarse evaluation is a lower bound for the fine one, and a positive coarse defect certifies a positive fine defect.*

6 A four-stage provenance experiment

We run a synthetic history through all four stages and report what the framework certifies. The experiment is in `provenance_demo.py`; all distributions are exact rationals.

Setup. There are three probes x_1, x_2, x_3 and six outcomes $y_1, \dots, y_6$. Corpus C_1 has two sources A_1, A_2, and corpus C_2 has one source N, which is the only one giving mass to y_6:

probe	A_1	A_2	N
x_1	$(.4, .3, .2, .1, 0, 0)$	$(0, .1, .2, .3, .4, 0)$	$(0, 0, 0, .2, .3, .5)$
x_2	$(.1, .1, .4, .4, 0, 0)$	$(.3, .3, .1, .1, .2, 0)$	$(0, 0, 0, 0, .5, .5)$
x_3	$(.5, .5, 0, 0, 0, 0)$	$(0, 0, .5, .5, 0, 0)$	$(0, 0, 0, 0, .5, .5)$

M_1 mixes A_1, A_2 with weights $(0.6, 0.4)$ at every probe. M_2 uses weights $(0.45, 0.25, 0.30)$ at x_1, $(0.5, 0.3, 0.2)$ at x_2, and leaves x_3 unchanged. The edit producing M_3 declares scope $\{x_1\}$ and moves 20% of the mass at x_1 to y_3, but also leaks 5% to y_1 at x_2. RLHF producing M_4 is a tilt with weights $(1, 2, 4, 1, 3, 1)$, so $K = 4$; a *drifted* variant additionally multiplies y_6 by 10 at x_2.

Stage 2: adding C_2. With exact sources the attribution is exact at every probe: the weights (A_1, A_2, N) are recovered as $(0.45, 0.25, 0.30)$, $(0.5, 0.3, 0.2)$ and $(0.6, 0.4, 0)$, with widths zero. Outcome y_6 appears at level $\varepsilon = 0.01$ at x_1 and x_2 and not at x_3, where $\mathrm{TV}(M_1, M_2) = 0$. Since y_6 is exclusive to N, proposition 3.7 gives $\lambda_N = p(y_6)/N(y_6)$, namely 3/10 and 1/5. An observation of y_5 at x_1 came from A_2 with probability 10/19 and from N with probability 9/19. The precision ladder at x_1 is:

stage	A_1	A_2	N	μ
exact sources, budget 0	0.45	0.25	0.30	0
supports only, budget 0	$[0.18, 0.66]$	$[0, 0.67]$	$[0.15, 0.52]$	0
supports only, budget 0.1	$[0.08, 0.66]$	$[0, 0.67]$	$[0.05, 0.52]$	$[0, 0.1]$

The structural imprecision is zero, the informational imprecision is large, and the budget lowers the lower ends further, as theorem 3.5 and proposition 3.6 predict. The lower bound 0.18 for A_1 is the mass of its exclusive outcome y_1.

Stage 3: the edit. The locality test with $\tau = 0.01$ passes at x_3 (TV $= 0$) and fails at x_2 ($\mathrm{TV}(M_2, M_3) = 0.043$), detecting the leak. Relative to the corpus sources A_1, A_2, N, the certified unmodeled weight is 0.20 at x_1 and 0.05 at x_2, exactly the mass moved by the edit and by the leak, with certificates y satisfying $\min_i \langle y, A_i \rangle = 1$ and $\langle y, p \rangle = 0.80$, respectively 0.95. At x_1 the total-variation change is 0.172, smaller than the certified unexplained weight 0.20: the two quantities answer different questions.

Stage 4: RLHF. For the pure tilt, the implied ratio is $K^* = 4$ at every probe, so every change is reachable with the claimed reward range. For the drifted variant, $K^* = 10$ at x_2, so the change is not reachable; the residual weight relative to the tilt class is 0.1934, with an exactly verified certificate. The other probes remain reachable, so the departure from the declared mechanism is localized. For the pure tilt at x_1, with $\beta = 1$ and claimed range $[0, \ln 4]$, the rewards are identified exactly, $r = (0, \ln 2, \ln 4, 0, \ln 3, 0)$, because $K^* = K$ (proposition 4.7); with a claimed range $[0, \ln 8]$ the reward differences are still exact, but each reward value has width $\ln 2$.

Paths. At x_1, with E the edit and R the tilt, the order defect is $\mathrm{TV}(REp, ERp) = 51/518 \approx 0.0985$, so the order matters. Tilting by w and then by $1/w$ has loop defect zero. Comparing E with a weaker edit E' (moving 25% instead of 20%), $\mathrm{TV}(REp, RE'p) = 0.0479$ is within the bound $\frac{3K-1}{2}\mathrm{TV}(Ep, E'p) = 5.5 \times 0.043$, with observed amplification 1.11. On the coarse evaluation $\{y_1, y_2\}, \{y_3, y_4\}, \{y_5, y_6\}$ the order defect is 0.0858, smaller but still positive, so the coarse evaluation alone certifies that the order matters.

Summary. The report separates what changed, what the declared sources and mechanisms explain, how precisely, and where they fail:

step	behavioral change	attribution	certified residual
add C_2	TV $= 0.18$ at x_1, x_2; 0 at x_3	exact; $\lambda_N = 0.3, 0.2, 0$	none
edit	TV $= 0.172$ at x_1; leak 0.043 at x_2	locality fails at x_2	0.20 (x_1), 0.05 (x_2)
RLHF	tilt, $K^* = 4$	reachable at all probes	none
RLHF, drifted	$K^* = 10$ at x_2	not reachable at x_2	0.19 (x_2)

7 The glue: 1-cells, 2-cells and assembly

The previous sections work probe by probe and compare interventions by numbers. This section supplies the structure that connects local states: *1-cells* that transport states and their attribution data, *2-cells* that witness how far two 1-cells differ, and *assembly*, which asks whether local states over several contexts come from one global state. We state exactly which laws hold strictly, which hold only as inequalities, and which fail. Every claim below is checked in `check_glue.py` (appendix A).

7.1 Kernel 1-cells

Definition 7.1 (Kernel 1-cells)**.** Objects are finite outcome sets. A *1-cell* $k\colon B \rightsquigarrow B'$ is a Markov kernel $k\colon B \to \mathsf{D}(B')$. It acts on states by Kleisli extension, $k^{\#}(p)(b') = \sum_b p(b)\, k(b)(b')$, and on

second-order states by $\mathsf{D}(k^{\#})$. 1-cells compose by $(h \circ k)(b) = h^{\#}(k(b))$, with identities the point masses η_B.

Coarsenings, restrictions of a joint outcome to some of its coordinates, noise channels and mixings $p \mapsto (1-a)p + a\,t$ (the kernel $b \mapsto (1-a)\delta_b + a\,t$) are 1-cells. Tilts are not: T_w is not affine in p, because of the normalization. They are Bayesian updates by the predicate w/K in the sense of [26], and they live in the Lipschitz layer of section 5, not in the Kleisli category.

Proposition 7.2 (Properties of 1-cells)**.** *For 1-cells $k\colon B \rightsquigarrow B'$ and $h\colon B' \rightsquigarrow B''$:*

(i) $k^{\#}$ *is affine;*

(ii) $(h \circ k)^{\#} = h^{\#}k^{\#}$ *and* $\eta_B^{\#} = \mathrm{id}$*;*

(iii) $k^{\#} \circ \mu = \mu \circ \mathsf{D}(k^{\#})$ *on second-order states, where μ is flattening;*

(iv) $\mathrm{TV}(k^{\#}p, k^{\#}q) \le \mathrm{TV}(p, q)$.

Proof. (i)–(iii) are the Kleisli-category laws of D, and (iii) says that $k^{\#}$ is a morphism of free D-algebras. (iv) is the data-processing inequality. □

Proposition 7.3 (Attribution is transported along 1-cells)**.** *Let $k\colon B \rightsquigarrow B'$. Transport each exact source A_i to $k^{\#}A_i$ and each support S_i to $k(S_i) = \bigcup_{b \in S_i} \operatorname{supp} k(b)$. A decomposition (s_i) of p with weights λ maps to the decomposition $(k_* s_i)$ of $k^{\#}p$ with the same weights. Hence*

$$\mu_{\min}(k^{\#}p) \le \mu_{\min}(p),$$

and for every budget, each weight interval at $k^{\#}p$ contains the corresponding interval at p: precision can only be lost along a 1-cell.

Proof. Pushforward k_* is linear and monotone on nonnegative measures and preserves mass, so $\sum_i k_* s_i \le k^{\#}p$, $k_*(\lambda_i A_i) = \lambda_i k^{\#}A_i$ and $\operatorname{supp} k_* s_i \subseteq k(S_i)$. Images of decompositions are decompositions with the same weights, which gives both claims. □

Remark 7.4 (Attribution as a fibre). With exact sources and no unmodeled part, the sources form a 1-cell $a\colon I \rightsquigarrow B$, $i \mapsto A_i$, from the index set of sources, and the attribution set is the fibre $(a^{\#})^{-1}(p) \subseteq \mathsf{D}(I)$. Precision is the size of this fibre, and proposition 7.3 is the inclusion $(a^{\#})^{-1}(p) \subseteq ((k \circ a)^{\#})^{-1}(k^{\#}p)$, which follows from $(k \circ a)^{\#} = k^{\#}a^{\#}$. In the demo, coarsening the six outcomes of x_1 into three classes lowers $\mu_{\min}$ at stage 3 from 0.20 to 0.029 and widens the weight intervals at budget 0.25 from $(0.05, 0.125, 0.10)$ to $(0.31, 0.62, 0.31)$.

7.2 Coupling 2-cells and their metric shadow

Definition 7.5 (Coupling 2-cells)**.** Let $f, g\colon X \rightsquigarrow Y$ be 1-cells. A *coupling 2-cell* $\alpha\colon f \Rightarrow g$ is a kernel $\alpha\colon X \to \mathsf{D}(Y \times Y)$ such that $\alpha(x)$ has first marginal $f(x)$ and second marginal $g(x)$ for every x. Its *cost* is $c(\alpha) = \max_x \Pr_{\alpha(x)}(y \neq y')$. The identity on f is the diagonal coupling $\Delta_f(x) = \sum_y f(x)(y)\,\delta_{(y,y)}$. The *vertical composite* of $\alpha\colon f \Rightarrow g$ and $\beta\colon g \Rightarrow h$ glues along the common marginal and then forgets it:

$$(\beta \cdot \alpha)(x)(y, y'') = \sum_{y'\colon g(x)(y') > 0} \frac{\alpha(x)(y, y')\,\beta(x)(y', y'')}{g(x)(y')}.$$

A coupling is a witness: it pairs the outputs of f and g and shows on which outputs they disagree. The gluing formula is the gluing lemma for couplings [43, Ch. 1].

Theorem 7.6 (The witness layer)**.**

(i) For fixed X, Y, 1-cells $X \rightsquigarrow Y$ and coupling 2-cells form a category: vertical composition has the right marginals, is associative, and the diagonal couplings are identities.

(ii) Cost is subadditive, $c(\beta \cdot \alpha) \le c(\alpha) + c(\beta)$, and $c(\Delta_f) = 0$.

(iii) $\min_\alpha c(\alpha) = d(f, g) := \max_x \mathrm{TV}(f(x), g(x))$, attained by the maximal coupling at each x [31].

(iv) Whiskering by a deterministic map $u\colon W \to X$, $\alpha \mapsto \alpha \circ u$, is strictly functorial.

(v) Whiskering by a stochastic 1-cell $u\colon W \rightsquigarrow X$ (mixing the couplings) and whiskering by a 1-cell $h\colon Y \rightsquigarrow Z$ (copying h on equal pairs and running it independently on unequal pairs) preserve identities and marginals and do not increase cost, but neither is functorial for vertical composition, even when h is deterministic.

Proof. (i) Fix x and drop it from the notation. For $\gamma\colon h \Rightarrow k$, both bracketings of the triple composite equal $\sum_{y',y''} \alpha(y,y')\beta(y',y'')\gamma(y'',y''')/(g(y')h(y''))$, the sum running over the supports of g and h. Marginals: summing over y'' gives $\sum_{y'} \alpha(y,y') = f(x)(y)$, and symmetrically. For the identity, $(\alpha \cdot \Delta_f)(y,y'') = f(y)\,\alpha(y,y'')/f(y)$. (ii) In the glued triple, $y \neq y''$ implies $y \neq y'$ or $y' \neq y''$. (iii) This is the coupling characterization of total variation. (iv) Both sides evaluate the same couplings at $u(w)$. (v) Take $X = \{x_0, x_1\}$, $Y = \{0,1\}$, all three 1-cells constant and uniform, and $\alpha = \beta$ equal to the diagonal at x_0 and to the swap coupling at x_1. Then $\beta \cdot \alpha$ is the diagonal at both points, and mixing it with weights $(\frac{1}{2}, \frac{1}{2})$ gives cost 0, whereas mixing first gives the independent coupling at each step, whose composite has cost $\frac{1}{2}$. For postcomposition, let h forget its input, $h(y) = \nu$ uniform: since swap $\cdot$ swap $= \Delta$, the whiskered composite has cost 0, while the composite of the whiskered swaps (each the independent coupling of ν with itself) has cost $\frac{1}{2}$. A deterministic h merging two of three outcomes also fails; the script exhibits an instance. The remaining claims are direct. □

So the witness layer has genuine hom-categories, whose composition is gluing, but the natural whiskerings are not functorial, and therefore it is not a 2-category or a bicategory with these operations. What survives is the inequality between costs, and that is exactly what the next result packages strictly.

Definition 7.7 (Graded 2-cells)**.** For 1-cells $f, g\colon X \rightsquigarrow Y$, a *graded 2-cell* $f \Rightarrow_\varepsilon g$ is a number $\varepsilon \ge d(f,g)$. Vertical composition adds grades, the identity on f has grade 0, and the horizontal composite of $f \Rightarrow_\varepsilon g$ (with $f, g\colon X \rightsquigarrow Y$) and $h \Rightarrow_\delta h'$ (with $h, h'\colon Y \rightsquigarrow Z$) is $hf \Rightarrow_{\delta+\varepsilon} h'g$.

Theorem 7.8 (The graded shadow is a strict 2-category)**.** *Finite sets, kernel 1-cells and graded 2-cells form a strict 2-category. Every coupling 2-cell $\alpha\colon f \Rightarrow g$ defines the graded 2-cell $f \Rightarrow_{c(\alpha)} g$; identities go to identities, and vertical composites go to grades no larger than the sum. The hom-pseudometric d makes kernel 1-cells a category enriched in Lawvere metric spaces.*

Proof. Validity of the composites: vertical by the triangle inequality, horizontal by

$$d(hf, h'g) \le d(hf, h'f) + d(h'f, h'g) \le d(h, h') + d(f, g),$$

using convexity of total variation for the first term and proposition 7.2(iv) for the second. Since grades compose by addition in a commutative monoid, associativity, unit laws, functoriality of horizontal composition and the interchange law $(\delta' + \delta) + (\varepsilon' + \varepsilon) = (\delta' + \varepsilon') + (\delta + \varepsilon)$ all hold. The statements about couplings are theorem 7.6(ii),(iii). □

Remark 7.9 (Why tilts break the strict shadow). For an L-Lipschitz map h' the horizontal grade becomes $\delta + L_{h'}\varepsilon$ (theorem 5.5(ii)). Interchange then compares $L_{h''}(\varepsilon+\varepsilon')$ with $L_{h''}\varepsilon' + L_{h'}\varepsilon$, which agree only if $L_{h'} = L_{h''}$. This is the 2-categorical reading of corollary 5.7: the strict structure holds for non-expansive maps, and tilts enter with Lipschitz weights.

Graded and ordered 2-cells. The graded shadow is enriched in Lawvere metric spaces: a 2-cell is a bound on a distance. A decision layer on credal sets uses the posetal counterpart, where a 2-cell between two transport laws is an inclusion of the sets they produce, and floor summaries are then a normal oplax functor [9, Prop. 10.24]. Both are thin, so their coherence is automatic; neither rescues the witness layer, whose whiskerings remain non-functorial. The typed obstruction record of that layer (a floor shortfall, a Farkas vector, a holonomy, a conflict among sources) keeps the attribution sets of this paper as one of its kinds of witness [48, Section 14].

7.3 Assembly: gluing local states

Definition 7.10 (Contexts and global sections [1])**.** Let V be a finite set of observables with outcome sets O_v, and let $\mathcal{C}$ be a cover of V by *contexts*. For $C' \subseteq C$ the restriction $r^C_{C'} \colon O^C \to O^{C'}$ is a deterministic 1-cell. An *empirical model* is a family $e = (e_C \in \mathsf{D}(O^C))_{C\in\mathcal{C}}$; it is *compatible* if e_C and $e_{C'}$ restrict to the same distribution on $C \cap C'$. A *global section* is $g \in \mathsf{D}(O^V)$ with $(r^V_C)^{\#} g = e_C$ for every C.

In provenance terms, an observable is a probe, and a context is a set of probes whose outputs are recorded jointly, for instance several questions answered in one conversation, or one evaluation item scored on several criteria. Per-probe evaluation is the case of singleton contexts, in which every family glues. We call gluing along restriction 1-cells *1-cell gluing*; it is an exact condition. *2-cell gluing* is gluing of witnesses, and assembly up to a graded 2-cell.

Proposition 7.11 (Two contexts: 1-cell gluing is 2-cell composition)**.** *Let C_1, C_2 be contexts with $I = C_1 \cap C_2$, and let e_1, e_2 be compatible with common restriction e_I. Then*

$$g(a,i,c) = \frac{e_1(a,i)\, e_2(i,c)}{e_I(i)} \qquad (a \in O^{C_1\setminus I},\ i \in O^I,\ c \in O^{C_2\setminus I})$$

(read as 0 when $e_I(i) = 0$) is a global section. Regarding e_1 as a coupling of its $C_1 \setminus I$ marginal with e_I, and e_2 as a coupling of e_I with its $C_2 \setminus I$ marginal, the marginal of g that forgets I is the vertical composite $e_2 \cdot e_1$ of definition 7.5.

Proof. Summing g over c gives e_1, and over a gives e_2. Forgetting i gives the gluing formula of definition 7.5. □

For more than two contexts, compatibility is not enough: a classical theorem of Vorob'ev states that every compatible family over a cover extends to a global section precisely when the cover is acyclic, in the sense that its contexts can be ordered so that each one meets the union of its predecessors inside a single predecessor [44]. The four contexts of the Clauser–Horne–Shimony–Holt (CHSH) scenario form a cycle, and the Popescu–Rohrlich (PR) box is compatible on every overlap without having a global section (appendix A). The next theorem measures the failure with the machinery of section 3.

Theorem 7.12 (Contextual fraction is minimal unmodeled weight)**.** *Let e be a compatible empirical model with $m = |\mathcal{C}|$ contexts. Put $p_e = \frac{1}{m}(e_C)_{C\in\mathcal{C}}$, a distribution on the disjoint union $\bigsqcup_C O^C$, and, for each deterministic global assignment $s \in O^V$, declare the exact source $A_s = \frac{1}{m}(\delta_{s|_C})_{C\in\mathcal{C}}$. Then*

$$\mu_{\min}(p_e) = \mathrm{CF}(e),$$

the contextual fraction of Abramsky, Barbosa and Mansfield [2]. In particular e has a global section if and only if $\mu_{\min}(p_e) = 0$, and a certificate y of theorem 3.2(iv) is an inequality $\langle y, A_s\rangle \geq 1$ satisfied by every global assignment and violated by e whenever $\langle y, p_e\rangle < 1$.

Proof. Mixtures $\sum_s \lambda_s A_s$ with $\sum_s \lambda_s = 1$ are exactly the vectors p_g of the restriction families of global distributions $g = \sum_s \lambda_s \delta_s$. In a decomposition $p_e = \sum_s \lambda_s A_s + \mu U$, every context block of $\sum_s \lambda_s A_s$ has mass $(1-\mu)/m$, so every block of μU has mass μ/m, and m times each block of U is a distribution e'_C; the family e' is compatible because e and the restriction family of g are. Thus decompositions are exactly the expressions $e = (1-\mu)e^{\mathrm{NC}} + \mu e'$ with e^{NC} noncontextual, and minimizing μ is the linear program defining CF in [2]. □

For the PR box mixed with uniform noise at visibility v, the linear program gives $\mathrm{CF} = \max(0, 2v-1)$ at $v \in \{1, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}\}$, with exactly verified certificates. The same construction with *declared* sources in place of deterministic assignments asks a provenance question.

Example 7.13 (Prompt-independent attribution)**.** At stage 2 of section 6, the corpus weights are identified exactly at each probe, with $\mu_{\min} = 0$ probe by probe, but they differ across probes. Declaring the sources A_1, A_2, N on all probes jointly and asking for one weight vector, as in theorem 7.12, gives $\mu_{\min} = \frac{1}{10}$ on $\{x_1, x_2\}$ and $\frac{3}{10}$ on $\{x_1, x_2, x_3\}$, both certified exactly. So no prompt-independent mixture of the declared corpora explains the joint behaviour, and $\frac{3}{10}$ is the least mass that any such explanation leaves unexplained. The local answers glue only if the explanation is allowed to depend on the prompt.

Definition 7.14 (Approximate gluing)**.** For any empirical model e, compatible or not,

$$\varepsilon^*(e) = \min_{g\in \mathsf{D}(O^V)} \max_{C\in\mathcal{C}} \mathrm{TV}\big((r_C^V)^{\#} g,\ e_C\big).$$

A minimizer is a global state that agrees with every context up to the graded 2-cell of grade ε^*.

Proposition 7.15 (Bounds for approximate gluing)**.**

(i) $\varepsilon^*(e)$ is the value of a linear program, and it is attained.

(ii) $\varepsilon^*(e) \geq \frac{1}{2}\max_{C,C'} \mathrm{TV}\big(e_C|_{C\cap C'},\ e_{C'}|_{C\cap C'}\big)$.

(iii) If e is compatible, $\varepsilon^*(e) \leq \mathrm{CF}(e)$, and $\varepsilon^*(e) = 0$ if and only if e has a global section.

(iv) For the PR box, $\varepsilon^* = \frac{1}{4}$, whereas $\mathrm{CF} = 1$.

Proof. (i) Introduce slack variables for the absolute values. (ii) For any g, by the triangle inequality and data processing, $\mathrm{TV}(e_C|_I, e_{C'}|_I) \leq \mathrm{TV}(e_C, g|_C) + \mathrm{TV}(g|_{C'}, e_{C'}) \leq 2\max_C \mathrm{TV}(g|_C, e_C)$. (iii) Write $e = (1-\kappa)e^{\mathrm{NC}} + \kappa e'$ with $\kappa = \mathrm{CF}(e)$ and let g be a global section of e^{NC}. Then $\mathrm{TV}(g|_C, e_C) = \kappa\,\mathrm{TV}(e^{\mathrm{NC}}_C, e'_C) \leq \kappa$. The equivalence follows since $\varepsilon^* = 0$ means that a global section exists. (iv) The PR box in context (i,j) is uniform on the outcomes with $o_1 \oplus o_2 = ij$. Every deterministic assignment violates this parity in an odd number of the four contexts, since the four left-hand sides sum to 0 and the right-hand sides to 1 modulo 2. Hence for any g the expected number of violated contexts is at least 1, some context is violated with probability at least $\frac{1}{4}$, and $\mathrm{TV}(g|_C, e_C)$ is at least that probability. The uniform distribution on the eight assignments with exactly one violation attains $\frac{1}{4}$ in every context. □

For the noisy PR family the linear program gives $\varepsilon^* = \frac{1}{4}\mathrm{CF}$ at $v = \frac{3}{4}$ and $v = \frac{9}{10}$, so the bound in (iii) can be loose by a factor of four. For the parity horns of the coupling nerve, $\varepsilon^* = \frac{1}{6}, \frac{3}{16}, 0.225, 0.25$ for $n = 3, 4, 5, 6$, against $\mathrm{CF} = 1 - 2^{2-n}$ [7, §13.1]. The two numbers answer different questions. CF is the weight that must be attributed to something outside every global explanation, while ε^* is how far the best single global state is from the worst context.

Gluing at level 2. The tower's second level carries uncertainty about output distributions, for instance a posterior over a model's predictive distribution. Gluing there is finer than gluing the flattened states.

Proposition 7.16 (Two-context gluing at level 2)**.** *Let $P_1 \in \mathsf{D}^2(O^{C_1})$ and $P_2 \in \mathsf{D}^2(O^{C_2})$ have finite support and the same image $Q = \mathsf{D}((r_I)^{\#})P_1 = \mathsf{D}((r_I)^{\#})P_2$ on the overlap I. Define*

$$G = \sum_q Q(q) \sum_{p_1, p_2} P_1(p_1 \mid q)\, P_2(p_2 \mid q)\, \delta_{p_1 \otimes_q p_2},$$

where $p_1 \otimes_q p_2$ is the gluing of proposition 7.11. Then G restricts to P_1 and to P_2, and its flattening μG is a global section of the pair $(\mu P_1, \mu P_2)$. In general $\mu G \neq \mu P_1 \otimes \mu P_2$: the level-2 gluing keeps a correlation between the contexts that runs through the uncertainty about the overlap.

Proof. Each $p_1 \otimes_q p_2$ restricts to p_1 and p_2 by proposition 7.11, and G is the gluing of P_1 and P_2 along Q, so its restrictions are P_1 and P_2. For the flattening, $(r_{C_1})^{\#}\mu G = \mu\, \mathsf{D}((r_{C_1})^{\#})G = \mu P_1$ by proposition 7.2(iii). The script gives an instance in which $\mathrm{TV}(\mu G, \mu P_1 \otimes \mu P_2) = 0.015$. □

8 Architecture: how the pieces fit, and why

8.1 The layers

Layer	Cells	Structure and status	Where
Ambient	finite sets, D	probability monad; mixing is its algebra structure	[9]
Local	per base B: context comonad, mixed law, admissibility	comonad with a strict mixed distributive law; obstruction tower	[9, §§9–10]
1-cells	kernels $k\colon B \rightsquigarrow B'$	Kleisli category; affine, compatible with flattening, non-expansive; transports attribution	propositions 7.2 and 7.3
2-cells (witness)	couplings	hom-categories by gluing (strict); whiskering not functorial, but a two-component A_∞-functor up to coherent homotopy	theorem 7.6; [6, Thm. 5.2]
2-cells (shadow)	grades $\varepsilon \geq d(f, g)$	strict 2-category; Lawvere-enriched; Lipschitz weights for tilts	theorems 5.5 and 7.8
Assembly	compatible families, global sections	exact for two contexts and acyclic covers; $\mathrm{CF} = \mu_{\min}$; approximate gluing ε^*; level-2 gluing	propositions 7.11, 7.15 and 7.16 and theorem 7.12
Nerve	couplings of all arities	simplicial object $\mathsf{D}(Y^{\bullet+1})$: 2-horns fill, higher horns are obstructed by CF	[6, §4]
Geometry	simplex as moment polytope; coupling polytopes	Fisher metric is round under $\sqrt{p}$; observation forgets Lagrangian phases; nondegenerate coupling polytopes are Delzant	proposition 8.1; [6, §6]

Why each layer is needed. The monad is needed because attribution *is* mixture decomposition: a decomposition is a preimage under the algebra structure of D. The local comonad is needed because what counts as an admissible belief depends on context, which is the subject of [9]. 1-cells are needed because every stage of a model's history (coarsening an evaluation, restricting to a context, mixing in data) must carry attribution data from one state to the next, and proposition 7.3 says this transport never manufactures precision. 2-cells are needed because "these two paths differ by 0.0985" (section 6) is a claim about a pair of 1-cells. The shadow composes such claims strictly, and a witness shows *where* the paths disagree, as a joint distribution of their outputs. Assembly is needed because real evaluations record several probes jointly, and example 7.13 shows that exact local attributions need not come from any single global explanation. The geometry is needed for the metric, and it is where a Fukaya-type structure would enter.

What is categorical here. A reader may ask why a paper about 2-categorical glue spends most of its pages on polytopes and linear programs. The answer is that the linear programs *compute* categorical objects:

- an attribution set is a fibre of the Kleisli extension of a 1-cell (the remark after proposition 7.3);
- the minimal unmodeled weight of a multi-context state is the failure of the sheaf condition (theorem 7.12);
- a reach set is the image of a family of maps of observable states, which for tilts are Bayesian predicate updates [26];
- comparisons are the hom-metric of a Lawvere-enriched category and the grades of a strict 2-category (theorem 7.8);
- witnesses are couplings, and their composition is the gluing lemma (theorem 7.6).

Category theory fixes what the objects are and how they compose. Convex geometry and duality are how one computes them on a finite probe set, with certificates. Neither replaces the other.

What is strict and what is not. Kernels form a strict category, the witness hom-categories are strict, and the graded shadow is a strict 2-category. The witness layer with its natural whiskerings is not a 2-category: functoriality fails, with explicit counterexamples, and only the cost inequality survives. Up to coherent homotopy the failure is organized: in the square-zero cone of the circulations, whiskering is an A_∞-functor with exactly two components, the second being the whiskering defect [6, Thm. 5.2]. No action of a stochastic kernel on couplings is both strictly unital and strictly compatible with composing kernels [6, Thm. 10.2]. Tilts are not kernels, and with them the shadow keeps its composition rule but loses strict interchange. Exact gluing holds for two contexts and for acyclic covers, and fails on cycles, where CF and ε^* measure the failure.

8.2 Geometry and the Fukaya programme

Proposition 8.1 (The simplex as a moment polytope)**.**

(i) The map $p \mapsto \sqrt{p}$ pulls the round metric of the unit sphere back to one quarter of the Fisher metric: $\sum_b v_b^2/p_b = 4\sum_b \big(v_b/(2\sqrt{p_b})\big)^2$. Consequently, the Fisher–Rao distance is $2\arccos\sum_b \sqrt{p_b q_b}$ [46, 4].

(ii) The map $[\psi] \mapsto (|\psi_b|^2/\|\psi\|^2)_b$ is the moment map of the standard torus action on $\mathbb{CP}^{n-1}$ with the Fubini–Study form, and its image is the simplex $\mathsf{D}(B)$ with $|B| = n$. The fibre over an interior p is a Lagrangian torus T^{n-1}, the set of amplitudes with $|\psi_b|^2 = p_b$, which differ only in their phases.

Proof. (i) is a direct computation; the distance follows because $\sqrt{p}$ lies on the unit sphere, where the geodesic distance is $\arccos\langle\sqrt{p}, \sqrt{q}\rangle$. (ii) is standard toric geometry [15]. Isotropy of the fibre is checked in `check_glue.py` in an affine chart of $\mathbb{CP}^2$: the Fubini–Study form vanishes on pairs of phase-rotation vectors. □

So an output distribution is a point of a moment polytope, and the information that observation discards (the phases of an amplitude realizing it) is a Lagrangian torus. This is the precise sense in which symplectic geometry sits under the simplex. It is a geometric fact about the ambient space, and nothing in this paper depends on it.

What the Fukaya category of $\mathbb{CP}^{n-1}$ sees. The objects of a Fukaya category are Lagrangians with brane data, its morphisms are Floer cochains, and its compositions count holomorphic polygons, so that composition is defined by gluing and is associative only up to coherent higher homotopies. For the torus fibres of $\mathbb{CP}^{n-1}$ the answer is sharp. A fibre with a flat $U(1)$ connection has nonvanishing Floer cohomology exactly when it lies over a critical point of the disc potential [13, 20]. For $\mathbb{CP}^{n-1}$ the only such fibre is the Clifford torus over the barycentre, whose Floer cohomology Cho computed [12]. In statistical terms, only the uniform distribution gives a nonzero object. We know of no statistical meaning for this fact, and we conclude that the ambient $\mathbb{CP}^{n-1}$ is not the right symplectic manifold for attribution. The companion theory [6] replaces it by the toric manifolds whose moment polytopes are the coupling polytopes of section 7.2, the spaces of witnesses. There the answer differs. The fibre over the product coupling is Floer-trivial for generic margins, which is proved; at the sampled points the Floer-nontrivial fibres lie over the leximin coupling, which is computed [6, Prop. 10.11, §10.4].

Necessary conditions for a genuine connection. We state the programme as conditions that any construction would have to meet. This paper does not prove any of them; the status of each, as decided elsewhere, is noted after it.

(F1) *A manifold built from the data.* Attribution sets and bounded-reward reach sets are polytopes (theorems 3.2 and 4.4), with rational facet normals when the data are rational. A compact polytope is the moment image of a compact toric symplectic manifold if and only if it is Delzant [15], and of a toric orbifold if and only if it is simple and rational [30]. When the condition holds, precision widths become lengths in a moment polytope and torus fibres sit over individual explanations. Attribution polytopes need not be simple, so this already restricts the cases covered. *Status:* for witness polytopes, the coupling polytopes $\mathrm{Cpl}(f(x), g(x))$, it holds whenever the margins are nondegenerate, and fails for the uniform 3×3 case, which is not simple [6, Thm. 6.4]. For attribution polytopes it is open.

(F2) *Higher compositions for witnesses.* Theorem 7.6(v) shows that the witness layer composes only up to an inequality. A Fukaya-type structure would need these failures to be organized as higher operations satisfying the A_∞ relations. *Status:* settled in the minimal model. The defects are the second Boolean cumulants of whiskering and form a two-component A_∞-functor into the square-zero cone, also across objects [6, Thm. 5.2, Thm. 10.2]. Larger models are open.

(F3) *Assembly as composition.* Proposition 7.11 identifies two-context gluing with vertical composition of witnesses. A geometric realization should send this identification to composition of Lagrangian morphisms, and send the obstruction measured by CF and ε^* on cyclic covers to an obstruction to composition. *Status:* the combinatorial half holds. Two-context gluing is the canonical filler of a 2-horn of the coupling nerve, and cyclic covers give higher horns whose filling is obstructed exactly by CF [6, Thm. 4.2, Thm. 4.3]. The Lagrangian half is a conjecture [6, Conj. 9.5], with a computed probe in [6, §10.4].

Why pursue it. The two features that distinguish the witness layer from a plain metric, composition by gluing and laws that hold only up to controlled error, are also the defining features of Fukaya categories. That is an analogy, not a theorem. Its value is that (F1)–(F3) are checkable: each is a finite statement about specific polytopes, couplings and covers, and parts of (F1)–(F3) have now been checked.

9 Relation to prior work

Most individual ingredients of this paper are known; the contribution is their combination into a calculus for model provenance whose reported object is a set of explanations together with its width and a certificate. We state the relation area by area and then summarize.

Mixture proportion estimation and contaminated data. Mixture proportion estimation studies $F = (1-\kappa)G + \kappa H$ with H known and G unknown; the maximal proportion $\kappa^*(F|H) = \operatorname{ess\,inf} f/h$ is identified, and κ itself is identified under irreducibility [10, 41], with extensions beyond irreducibility [47]. Several known components plus one unknown appear in class-proportion estimation with anomaly rejection [39], in open-set label shift, where identifiability needs positivity or separability conditions [21], and in mutual contamination models [11]. In econometrics, Horowitz and Manski derive sharp bounds under contaminated sampling with a known bound on the error probability [22], within the program of partial identification [32]. Our minimal unmodeled weight is the finite-outcome complement of κ^* (remark 3.3), and our budget is the Horowitz–Manski error bound. These works estimate proportions from samples under identifying assumptions. We instead treat output distributions as known, characterize the *whole* set of decompositions whether or not it is identified, allow sources known exactly and sources known only by their supports in the same model, report sharp intervals for every weight and every observation-level attribution, and attach exact certificates.

Imprecise probability. Attribution sets are credal sets of decompositions in the sense of imprecise probability [45]; for sources known by their supports, the weight intervals are belief and plausibility of Dempster–Shafer theory [16, 42] ([9, Prop. 5.8]). The density ratio classes of robust Bayesian analysis [17] are exactly the reach sets of bounded-reward RLHF (remark 4.5). What is new here is the precision ladder that separates structural, informational and unmodeled imprecision (theorem 3.5), and the fact that slack in the unmodeled budget always costs precision (proposition 3.6).

RLHF. The Gibbs form of the optimum of KL-regularized reinforcement learning is known [28, 37]; we use it. The new point is a membership test for the resulting class of changes and a certified residual when a change falls outside it (theorems 4.4 and 4.9).

Training-data attribution. Influence functions [27], datamodels [25] and TRAK [36] estimate how individual training examples affect predictions, approximating counterfactual retraining. They answer a different question. We take sources as given, for instance reference models trained on each corpus, and certify what observed output distributions do and do not determine about them. The two are complementary: data-attribution estimates can be used to construct declared sources.

Model editing, diffing and composition. Locality or specificity of weight edits is measured as an evaluation metric [34], and edits are known to have ripple effects beyond their intended scope [14]. Here a locality claim is a mechanism class, and its violation comes with a certified residual (example 4.2, theorem 4.9). Crosscoder-based model diffing compares the internal features of a base and a fine-tuned model [35]; our comparisons are behavioral, at the probe outputs. Task arithmetic composes interventions as vectors in weight space [24]; the order defect measures whether composition commutes in the outputs.

Metric semantics. Categories enriched in generalized metric spaces go back to Lawvere [29]; behavioral pseudometrics for probabilistic systems [18], quantitative algebraic reasoning [33] and sensitivity type systems [38] all use composition rules with Lipschitz factors. Theorem 5.5 is an instance. What is specific here is that interventions act on model states and are compared only through a non-faithful observation map, which can rule out any Lipschitz constant (proposition 5.4), and the explicit constants for tilts (proposition 5.3).

Contextuality and comonadic simulation. The relation to contextuality is one of extension, not replacement. Abramsky and Brandenburger formulate an empirical model as compatible local probability data and characterize contextuality by the existence of a global section [1]. Abramsky, Barbosa, Karvonen and Mansfield then gave a comonadic treatment of simulations, that is, of morphisms between empirical models [3]. Morphisms are therefore not our novelty. Our local object is richer: for each base B it contains the ambient probability structure together with a base-dependent context comonad, a mixed distributive law, higher-order admissibility data and an obstruction tower [9]. The maps considered here are meant to transport these whole local structures, and the present paper equips such transport with quantitative comparison and path defects. For the global problem, theorem 7.12 shows that the contextual fraction of Abramsky, Barbosa and Mansfield [2] is the minimal unmodeled weight of section 3 for a particular choice of sources, namely the deterministic global assignments. Declaring reference models as sources instead turns the same linear program into a test for prompt-independent attribution (example 7.13). Abramsky and Brandenburger work with sections and do not use 2-cells. Here two-context gluing is identified with vertical composition of coupling 2-cells (proposition 7.11), approximate assembly is graded by the shadow (proposition 7.15), and gluing is carried out at the second level of the tower (proposition 7.16). The extension criterion for acyclic covers is Vorob'ev's [44].

Couplings, information geometry and toric geometry. The gluing lemma and the coupling characterization of total variation are standard [43, 31]. Our use of them is to organize witnesses of comparisons into hom-categories and to show where functoriality fails (theorem 7.6). Bayesian updating as predicate transformation is due to Jacobs and Zanasi [26]. The square-root embedding and the Fisher–Rao distance are classical [46, 4], and the moment map of $\mathbb{CP}^{n-1}$, Delzant's theorem and the Floer theory of toric fibres are standard [15, 30, 12, 13, 20]. Nothing in section 8.2 is a new theorem. What is new there is the list of conditions (F1)–(F3) that a Fukaya-type realization of

attribution would have to satisfy. The companion theory [6], with computations in [7], proves (F2) in the square-zero model and (F1) for witness polytopes, and reduces (F3) to its Lagrangian half.

Summary.

Result	Closest prior work	What is new here
Theorem 3.2	mixture proportion estimation; contaminated-sampling bounds	full polytope of decompositions for several exact or support-known sources; sharp weight and observation intervals; exact certificates
Theorem 3.5, proposition 3.6	credal sets; partial identification	monotone ladder of structural, informational and unmodeled imprecision; slack costs precision
Theorem 4.4	density ratio classes; Gibbs form of RLHF	RLHF reach is a density ratio class; implied reward range $\beta \log K^*$
Theorem 4.9	LP duality	residual of a change relative to a mechanism class, with certificate
Theorem 5.5, propositions 5.3 and 5.4	behavioral metrics; quantitative algebra; sensitivity typing	interventions seen through a non-faithful observation map; constants for tilts; no-Lipschitz obstruction
Proposition 7.3, theorems 7.6 and 7.8	Kleisli categories; gluing lemma; Lawvere metrics	attribution transported along 1-cells; witness hom-categories and explicit failures of whiskering; strict graded shadow
Theorem 7.12, example 7.13	contextual fraction [2]	CF $= \mu_{\min}$; the same program with declared sources tests prompt-independent attribution
Propositions 7.11, 7.15 and 7.16	sheaf-theoretic contextuality; Vorob'ev	gluing as 2-cell composition; bounds for approximate gluing, $\varepsilon^* = \frac{1}{4}$ for the PR box; level-2 gluing
Section 6	—	an end-to-end provenance report with widths and certificates

We do not claim as new the linear-programming and duality machinery, the single-source formula of mixture proportion estimation, the Gibbs form of RLHF, density ratio classes, or the Lipschitz composition calculus, the gluing lemma, the contextual fraction, or the geometry of proposition 8.1.

10 Scope and open problems

Local structure and glue. In the architecture of [9], the probability monad D is ambient, and each base B carries local structure: a context comonad, a distributive law, admissibility data and an obstruction tower. This paper supplies the transport between local structures: interventions, mechanism classes and quantitative comparison, and, in section 7, kernel 1-cells, coupling 2-cells and assembly. Deterministic base change is a structure-preserving map in [9, Prop. 9.13]. Kernel 1-cells act on the first two levels compatibly with flattening (proposition 7.2), and in fact on the full local structure: the context comonad, the mixed law, admissibility data and obstructions are transported along kernels, obstructions can only disappear, and certificates pull back [9, Prop. 9.14, Cor. 9.15]. The context functor is then a discrete opfibration over the Kleisli category [6, §3].

Structural provenance. The obstruction tower of [9, §10] gives a second coordinate. For a claim c attached to a model, let $k(c)$ be its first failing level. Along an intervention, the pair

$(k_{\text{before}}, k_{\text{after}})$ records which level of admissibility changed. Because failure is monotone up the tower [9, Thm. 10.5], a transition $k : 1 \to 2$ means that a level-one obstruction was removed while a level-two one remains; it does not mean that a level-two obstruction was created. We leave the systematic study of such transitions to future work.

Open problems.

1. *Finite-sample precision.* Output distributions are estimated from samples. Replacing p by a confidence region $\mathcal{C}_\delta$ turns each linear program into a robust program and each width into a statistical width. Two pieces exist: a finite-sample test built from a certificate [9, Prop. 11.1], and a sample bound for identifying the chamber of a pair of margins [6, Prop. 12.2].

2. *Weight-level metrics.* Choose a metric on $\mathcal{W}$ for which real interventions are Lipschitz, so that theorem 5.5 applies beyond output-level mechanisms.

3. *Higher levels.* Lift comparisons to second- and third-order states with the Kantorovich metric [19], and relate them to the obstruction tower. At level two the graded obstruction is a Kantorovich distance, and a claim obstructed there cannot learn [8, Lemma 52.1, Cor. 52.3]; level three is open.

4. *Cyclic covers at level 2.* Proposition 7.16 glues two contexts at level 2. For cyclic covers, define and compute a level-2 contextual fraction and compare it with the level-1 value of the flattened model.

5. *A functorial witness layer.* Find whiskerings of coupling 2-cells that are functorial, for example by indexing couplings by pairs of inputs as in relational program logics, or prove that none exist for noisy 1-cells. Up to coherent homotopy this is answered [6, Thm. 5.2]. Strictly, the whiskering that keeps diagonal couplings as identities does not compose along stochastic kernels, while the product action that does is not unital [6, Thm. 10.2]. Couplings with kernels as vertical maps form a double category only when those maps are bijections [5].

6. *Fukaya conditions.* Parts are decided (section 8.2). Decide (F1)–(F3) of section 8.2 for the attribution polytopes that arise in practice.

7. *Empirical validation.* Run the provenance report on real checkpoints, with declared sources from reference models.

A Code and computational checks

The library `provenance.py` implements attribution sets, widths, residuals and certificates. The script `check_attribution.py` verifies the following. theorem 3.2 (nonemptiness criterion, the interval $[\mu_{\min}, \bar{\mu}]$, and exactly verified dual certificates); theorem 3.5 (monotone widths along the ladder and in the budget); proposition 3.6; theorem 4.4 in exact arithmetic (every tilt satisfies the inequalities, every point satisfying them is reconstructed exactly as a tilt, and mixtures of tilts are tilts); proposition 4.6; theorem 4.9 for the tilt class (exactly verified certificates (y, ν), and zero residual exactly for reachable changes); proposition 5.3 (the exact constant for mixing, the tilt bound, and an example with amplification 4.0 for $K = 4$); proposition 5.4; theorem 5.5(i)–(iii) in exact arithmetic; proposition 5.9; proposition 4.7 (the reward interval reproduces the tilt at both endpoints, leaves the range just beyond them, and has width $\Delta_r - \beta \log K^*$); the observable reach of the example of proposition 5.4 has two points, so proposition 4.12 needs set-valued classes; and

corollary 4.13 (the residual relative to the enlarged class vanishes for perturbed tilts within ε and is positive outside). All 24 checks pass. The script `check_glue.py` verifies sections 7 and 8 in 30 checks. *1-cells:* proposition 7.2(i)–(iv) in exact arithmetic, and proposition 7.3 for a coarsening and a noisy coarsening of the demo state. *2-cells:* the marginals, associativity and identities of theorem 7.6(i) exactly, including couplings with zeros; subadditivity of cost; minimal cost equal to total variation by linear programming; strict functoriality for deterministic precomposition; the two counterexamples of theorem 7.6(v) and a counterexample for a deterministic merging map; the laws of theorem 7.8. *Assembly:* proposition 7.11; theorem 7.12 for the noisy PR family, with exactly verified certificates; compatibility of the PR box on all overlaps; proposition 7.16; proposition 7.15(ii)–(iv), with an exact feasible section and the parity lower bound for $\varepsilon^* = \frac{1}{4}$; example 7.13. *Geometry:* proposition 8.1, the Fisher–Rao distance by integrating along a great circle, and isotropy of the torus fibres in $\mathbb{CP}^2$. All 30 checks pass. The script `provenance_demo.py` produces the report of section 6. Linear programs are solved in floating point, and their certificates are rounded to rationals and verified exactly.

References

[1] S. Abramsky and A. Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011.

[2] S. Abramsky, R. S. Barbosa, and S. Mansfield. Contextual fraction as a measure of contextuality. *Physical Review Letters*, 119:050504, 2017.

[3] S. Abramsky, R. S. Barbosa, M. Karvonen, and S. Mansfield. A comonadic view of simulation and quantum resources. In *Proceedings of the 34th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*, 2019.

[4] S. Amari and H. Nagaoka. *Methods of Information Geometry*. Translations of Mathematical Monographs 191. American Mathematical Society and Oxford University Press, 2000.

[5] V. K. Awasthi. Coupling algebras: gluing, the square-zero cone and the ring of contingency tables. Technical report, Johns Hopkins University, October 2026. Draft 1. Code: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/coupling_algebra.

[6] V. K. Awasthi. Conditionals, couplings and toric geometry: kernels, the coupling nerve, cumulants and a Fukaya programme. Technical report, Johns Hopkins University, October 2026. Revised draft 4. Code: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/conditionals.

[7] V. K. Awasthi. Coupling polytopes and toric geometry: the 2×3 family, the square case, higher levels and stability. Companion paper, Johns Hopkins University, October 2026. Code: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/coupling.

[8] V. K. Awasthi. Three layers of realisation: hierarchy recovery, holonomy, information, contextuality and categorical dynamics. Technical report, Johns Hopkins University, September 2026, revised October 2026. DOI: 10.13140/RG.2.2.24274.52163. Code: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/AMB-Vigneaux/.

[9] V. K. Awasthi. The three-level posterior tower: hierarchical Bayesian inference and mixed distributive laws. Technical report, Johns Hopkins University, October 2026. Revised version 10. doi:10.13140/RG.2.2.29743.50088 (an earlier version, titled "... and weak distributive laws"). Code: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/Heirarchical_posteriors.

[10] G. Blanchard, G. Lee, and C. Scott. Semi-supervised novelty detection. *Journal of Machine Learning Research*, 11:2973–3009, 2010.

[11] G. Blanchard and C. Scott. Decontamination of mutually contaminated models. In *Proceedings of the 17th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 33 of *PMLR*, 2014.

[12] C.-H. Cho. Holomorphic discs, spin structures, and Floer cohomology of the Clifford torus. *International Mathematics Research Notices*, 2004(35):1803–1843, 2004.

[13] C.-H. Cho and Y.-G. Oh. Floer cohomology and disc instantons of Lagrangian torus fibers in Fano toric manifolds. *Asian Journal of Mathematics*, 10(4):773–814, 2006.

[14] R. Cohen, E. Biran, O. Yoran, A. Globerson, and M. Geva. Evaluating the ripple effects of knowledge editing in language models. *Transactions of the Association for Computational Linguistics*, 12, 2024.

[15] T. Delzant. Hamiltoniens périodiques et images convexes de l'application moment. *Bulletin de la Société Mathématique de France*, 116(3):315–339, 1988.

[16] A. P. Dempster. Upper and lower probabilities induced by a multivalued mapping. *Annals of Mathematical Statistics*, 38(2):325–339, 1967.

[17] L. DeRobertis and J. A. Hartigan. Bayesian inference using intervals of measures. *Annals of Statistics*, 9(2):235–244, 1981.

[18] J. Desharnais, V. Gupta, R. Jagadeesan, and P. Panangaden. Metrics for labelled Markov processes. *Theoretical Computer Science*, 318(3):323–354, 2004.

[19] T. Fritz and P. Perrone. A probability monad as the colimit of spaces of finite samples. *Theory and Applications of Categories*, 34(7):170–220, 2019.

[20] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono. Lagrangian Floer theory on compact toric manifolds I. *Duke Mathematical Journal*, 151(1):23–174, 2010.

[21] S. Garg, S. Balakrishnan, and Z. C. Lipton. Domain adaptation under open set label shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[22] J. L. Horowitz and C. F. Manski. Identification and robustness with contaminated and corrupted data. *Econometrica*, 63(2):281–302, 1995.

[23] P. J. Huber. Robust estimation of a location parameter. *Annals of Mathematical Statistics*, 35(1):73–101, 1964.

[24] G. Ilharco, M. T. Ribeiro, M. Wortsman, S. Gururangan, L. Schmidt, H. Hajishirzi, and A. Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations (ICLR)*, 2023.

[25] A. Ilyas, S. M. Park, L. Engstrom, G. Leclerc, and A. Madry. Datamodels: predicting predictions from training data. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.

[26] B. Jacobs and F. Zanasi. A predicate/state transformer semantics for Bayesian learning. *Electronic Notes in Theoretical Computer Science*, 325:185–200, 2016.

[27] P. W. Koh and P. Liang. Understanding black-box predictions via influence functions. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.

[28] T. Korbak, E. Perez, and C. L. Buckley. RL with KL penalties is better viewed as Bayesian inference. In *Findings of the Association for Computational Linguistics: EMNLP*, 2022.

[29] F. W. Lawvere. Metric spaces, generalized logic, and closed categories. *Rendiconti del Seminario Matematico e Fisico di Milano*, 43:135–166, 1973.

[30] E. Lerman and S. Tolman. Hamiltonian torus actions on symplectic orbifolds and toric varieties. *Transactions of the American Mathematical Society*, 349(10):4201–4230, 1997.

[31] T. Lindvall. *Lectures on the Coupling Method*. Wiley, 1992.

[32] C. F. Manski. *Partial Identification of Probability Distributions*. Springer, 2003.

[33] R. Mardare, P. Panangaden, and G. Plotkin. Quantitative algebraic reasoning. In *Proceedings of the 31st Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*, pages 700–709, 2016.

[34] K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[35] J. Minder, C. Dumas, C. Juang, B. Chughtai, and N. Nanda. Overcoming sparsity artifacts in crosscoders to interpret chat-tuning. arXiv:2504.02922, 2025.

[36] S. M. Park, K. Georgiev, A. Ilyas, G. Leclerc, and A. Madry. TRAK: attributing model behavior at scale. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.

[37] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn. Direct preference optimization: your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[38] J. Reed and B. C. Pierce. Distance makes the types grow stronger: a calculus for differential privacy. In *Proceedings of the 15th ACM SIGPLAN International Conference on Functional Programming (ICFP)*, 2010.

[39] T. Sanderson and C. Scott. Class proportion estimation with application to multiclass anomaly rejection. In *Proceedings of the 17th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 33 of *PMLR*, 2014.

[40] A. Schrijver. *Theory of Linear and Integer Programming*. Wiley, Chichester, 1986.

[41] C. Scott. A rate of convergence for mixture proportion estimation, with application to learning from noisy labels. In *Proceedings of the 18th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 38 of *PMLR*, 2015.

[42] G. Shafer. *A Mathematical Theory of Evidence*. Princeton University Press, 1976.

[43] C. Villani. *Optimal Transport: Old and New*. Grundlehren der mathematischen Wissenschaften 338. Springer, 2009.

[44] N. N. Vorob'ev. Consistent families of measures and their extensions. *Theory of Probability and Its Applications*, 7(2):147–163, 1962.

[45] P. Walley. *Statistical Reasoning with Imprecise Probabilities*. Chapman and Hall, 1991.

[46] W. K. Wootters. Statistical distance and Hilbert space. *Physical Review D*, 23(2):357–362, 1981.

[47] Y. Zhu, A. Fjeldsted, D. Holland, G. Landon, A. Lintereur, and C. Scott. Mixture proportion estimation beyond irreducibility. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, 2023.

[48] V. K. Awasthi, Empirical site separation: bending, tearing, contextuality and holonomy in a national survey, retrained language models and clinical data. Technical report, Johns Hopkins University, October 2026. Code: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/monitor.